

Group Project Report – Hire Wire

Ciara Slater – 8607497

Siew Tin Lee – 9496865

Nikita Veliev – 8773142

James Kennedy – 8654086

Qianran Xiu – 8788728

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Development Document

Project Overview (10%) - Nick

Project Background

The recruitment process is an important part of the modern job market. Job seekers look for suitable job opportunities, and employers look for suitable candidates. However, the recruitment process is often difficult and time consuming for both candidates and employers.

Many job search platforms provide many job postings and applications, but it is not always easy to find the best match. Job seekers often spend a lot of time searching for relevant jobs, and employers receive many applications from candidates who do not meet the job requirements. This makes the recruitment process inefficient and slow.

To solve this problem, the Intelligent Talent Matching Platform is proposed. This system is designed to match candidates and job opportunities based on their profiles, skills, education, and work experience. The system aims to make the recruitment process faster and more efficient. In this system, candidates can create profiles and upload their information, and employers can create job postings with specific requirements. The system then recommends the most relevant jobs to candidates and the most suitable candidates to employers.

Problem Definition

The biggest problem for candidates in the job market is that it is often very difficult to find a suitable job that fits their skills and expectations. Also, many talented candidates with strong resumes do not even get a chance to reach the first interview because automated AI filters may reject their resumes due to keyword issues.

As for companies, they receive many irrelevant applications from candidates who apply for many jobs without meeting the requirements. With such a high number of applications, it becomes very difficult to identify the most suitable candidates. As a result, companies may hire the wrong candidate or fail to fill the position successfully.

Without an intelligent matching system with effective filtering, the process of finding a job or a suitable candidate becomes inefficient and time-consuming. Therefore, a smart recommendation system is needed in today's highly competitive job market.

System Objectives

The main objectives of the system are:

- To develop a web-based recruitment platform for candidates and employers.
- To allow candidates to create and manage their profiles.
- To allow employers to create and manage job postings.
- To provide search and filtering functions for jobs and candidate profiles.
- To implement a recommendation system for job and candidate matching.
- To improve the efficiency of the recruitment process in general.

Target Users

The target users of the system include candidates, employers, and system administrators. Candidates use the system to create profiles, search for job opportunities, and receive job recommendations. Employers use the system to post job vacancies, search for candidates, and receive candidate recommendations. System administrators manage the platform, user accounts, and system data.

Requirements Analysis (15%) - James & Ciara

Analyse the Needs of Target Users and Customers

By analysing the needs of users, *Hire Wire* can get a better understanding of the program that needs to be developed and delivered, in order to meet the expectations of customers. *Hire Wire* has two main users/customers: employers and employees. Employers are customers that use *Hire Wire* to post job listings and match with employees, allowing them to expand their workforce. Employees are customers that are seeking a job and looking to match with employers that meet their needs. Employers and employees both have similar needs for *Hire Wire*:

1. **A user-friendly layout and navigation** – A user-friendly layout is a fundamental need due to the level of importance it holds within the website and overall user experience. Without a user-friendly layout and navigation, the user's experience becomes difficult and frustrating, making it extremely difficult to retain recurring customers.
2. **Efficient performance and processes** – Efficient performance is necessary to keep users engaged with the website. With slow processes and performance, unique aesthetics and intelligent features become meaningless.
3. **Intuitive and good usability** – Good usability creates a flourishing user environment in which customers can have a natural and reliable experience. Intuitive usability gives user's the ability to respond automatically to features within the website, enhancing the user instinctual experience.

4. **Helpful customer support** – With helpful customer support comes happy customers. If user's do not have anywhere to go for support or anyone to contact for issues that may arise within the website, friction can be created between the users and the developers.
5. **Safe and secure data handling** – Customers rely on *Hire Wire* to respect and handle their data safely and securely. Without giving customers transparency about how their data is stored, and how it is used, they can not feel safe about entrusting their data to *Hire Wire*.
6. **Memorable and learnable features** – For a website to be memorable, it needs features that are learnable and easy to remember. This includes creating features that are reliable, intuitive, and useful. Checking these boxes allows *Hire Wire* to leave a good impression and imprint on the user's memory.

Identify Major Functional Needs of the System

The employee(s) should be able to:

- See all jobs available on the site
- Search for jobs on the site with any appropriate filters that the user wants, using a keyword search
- Easily navigate the site
- See the matches that they make with employers
- Upload their resume or complete a candidate information form
- Create a profile based on either their resume or candidate form

The employer(s) should be able to:

- See which users/potential employees match with the employer's criteria for a given job listing
- Publish job listings, with appropriate requirements for candidates
- Search for users by either using a keyword search or filter
- Search for users based on certain criteria
- Intuitive and easy for all customers to use

The system needs to:

- Provide the two-way matching mechanism
- Provide recommended jobs to users/candidates, recommending the 10 most likely-to-be-relevant jobs
- Provide employers with the 10 most likely-to-be-successful candidates
- Apply appropriate recommendation algorithms to both users and employers

User Stories

- As a user, I want to stay logged in so that I don't have to enter my credentials every time every time I enter the website.

#6 Add feature to keep users logged in

USER STORY

Add tag +

Created by Clara Slater
29 Mar 2026 20:12

As a user, I want to stay logged in so that I don't have to enter my credentials every time I enter the website.

0 Attachments

Drop attachments here!

Tasks

0 Comments 7 Activities

Type a new comment here

OPEN NEW

POINTS

UX	1/2
Design	0
Front	0
Back	1
Product Owner	0
total points	1.5

ASSIGNED

James K

+ Add assigned Assign to me

WATCHERS

+ Add watchers Watch

🕒 👤 📁 🔒 🗑️

- As a user, I want to be able to reset my password so that I can regain access to my account if I forget my password.

#7 Add reset password feature

USER STORY

Add tag +

Created by Clara Slater
29 Mar 2026 20:13

As a user, I want to be able to reset my password so that I can regain access to my account if I forget my password.

0 Attachments

Drop attachments here!

Tasks

0 Comments 4 Activities

Type a new comment here

OPEN NEW

POINTS

UX	1/2
Design	0
Front	1/2
Back	1
Product Owner	0
total points	2

ASSIGNED

Nikita

+ Add assigned Assign to me

WATCHERS

+ Add watchers Watch

🕒 👤 📁 🔒 🗑️

- As a user, I want to browse all available job listings so that I can weigh my options.

#8 Create job listings page

USER STORY

Add tag +

Created by Clara Slater
29 Mar 2026 20:15

As a user, I want to browse all available job listings so that I can weigh my options.

0 Attachments

Drop attachments here!

Tasks

0 Comments 7 Activities

Type a new comment here

OPEN NEW

POINTS

UX	1
Design	3
Front	3
Back	5
Product Owner	?
total points	12

ASSIGNED

Clara Slater

+ Add assigned

WATCHERS

+ Add watchers Watch

🕒 👤 📁 🔒 🗑️

- As a user, I want to search for specific jobs/employees so that I can view jobs/employees that are more relevant to my specific wants and needs.

#9 Add filter search feature to job listings page
USER STORY

Created by Clara Slater
29 Mar 2026 20:16

Add tag +

As a user, I want to search for specific jobs/employees so that I can view jobs/employees that are more relevant to my specific wants and needs.

0 Attachments +
Drop attachments here!

Tasks +

0 Comments 4 Activities

Type a new comment here

POINTS

UX	1/2
Design	1/2
Front	1
Back	3
Product Owner	0
total points	5

ASSIGNED

Katie

+ Add assigned Assign to me

WATCHERS

+ Add watchers Watch

- As a user, I want to match with employees/employers so that I can view relevant candidates/apply for relevant positions.

#10 Matching system
USER STORY

Created by James K
30 Mar 2026 10:37

Add tag +

As a user, I want to match with employees/employers so that I can view relevant candidates/apply for relevant positions.

0 Attachments +
Drop attachments here!

Tasks +

0 Comments 4 Activities

POINTS

UX	2
Design	1
Front	1
Back	3
Product Owner	?
total points	7

ASSIGNED

+ Add assigned Assign to me

WATCHERS

+ Add watchers Watch

- As a user, I want to upload job listings with certain appropriate criteria so that potential employees can properly understand the job opening.

#11 Employer job listing upload
USER STORY

Created by James K
30 Mar 2026 10:42

Add tag +

As a user, I want to upload job listings with certain appropriate criteria so that potential employees can properly understand the job opening.

0 Attachments +
Drop attachments here!

Tasks +

0 Comments

POINTS

UX	1
Design	1
Front	1
Back	3
Product Owner	1/2
total points	6.5

ASSIGNED

+ Add assigned Assign to me

WATCHERS

+ Add watchers Watch

- As a user, I want to upload either my resume or complete a candidate information form so that I save time filling when applying for jobs.

#12 Candidate resume/form upload

USER STORY

Add tag +

Created by James K
30 Mar 2026 10:44

As a user, I want to upload either my resume or complete a candidate information form so that I save time filling when applying for jobs.

0 Attachments

Drop attachments here!

Tasks

OPEN NEW

POINTS

UX	2
Design	1
Front	1
Back	3
Product Owner	1
total points	8

ASSIGNED

+ Add assigned

Assign to me

WATCHERS

+ Add watchers

Watch

- As a user, I want to publish job listings so that relevant candidates can apply for positions.

#13 Employer job listing upload (more basic)

USER STORY

Add tag +

Created by James K
30 Mar 2026 10:58

As a user, I want to publish job listings so that relevant candidates can apply for positions.

0 Attachments

Drop attachments here!

Tasks

0 Comments

OPEN NEW

POINTS

UX	?
Design	?
Front	?
Back	?
Product Owner	?
total points	?

ASSIGNED

+ Add assigned

Assign to me

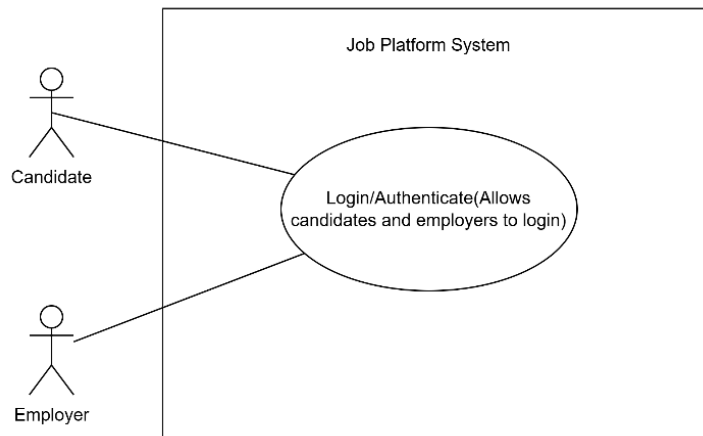
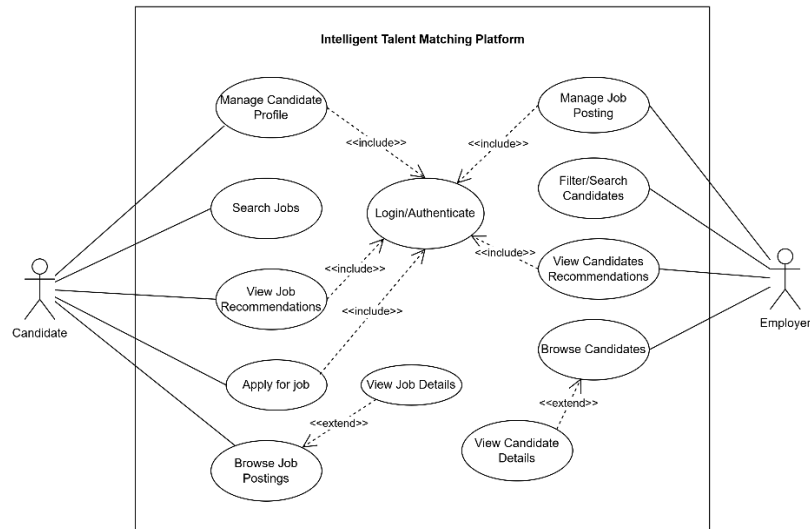
WATCHERS

+ Add watchers

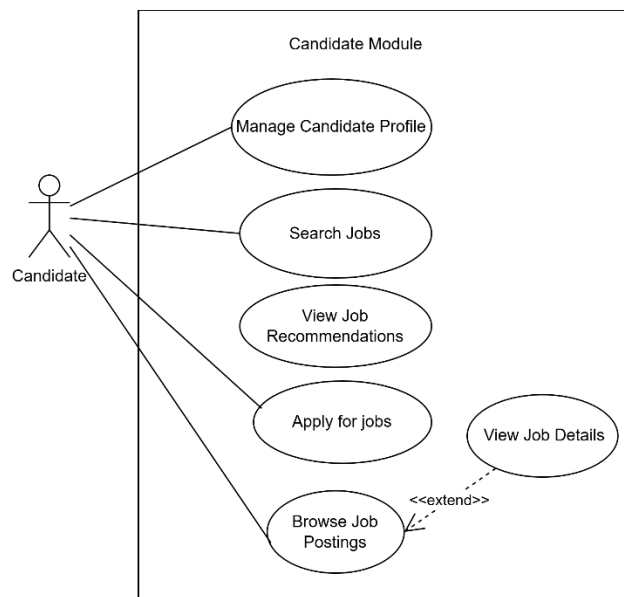
Watch

System Analysis & Design (15%) – Siew Tin & Qianran

Use Case Descriptions

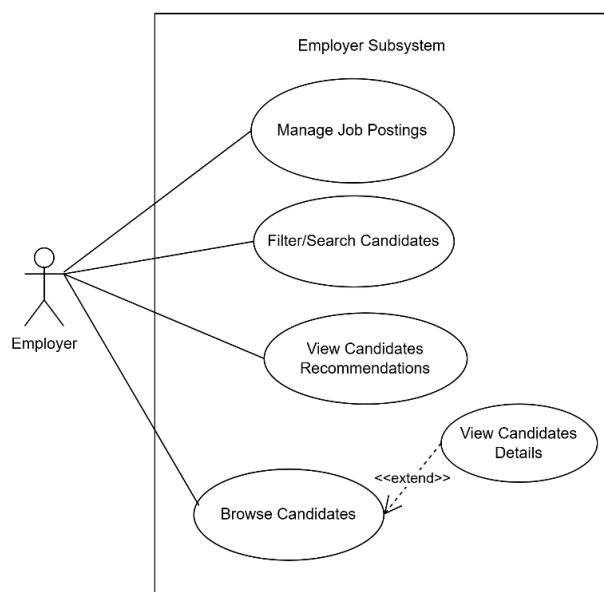


Use Case for Login/Authenticate	
Use case 1	Login/Authenticate
Actor	Candidate, Employer
Description	Allows candidates and employers to login with credentials to access the platform services
Basic Flow	1. User enters their username and password. 2. System validates the user's credentials. 3. If credentials are valid, user is granted access to their account. 4. User will be redirected to the homepage



Use Cases for Candidate	
Use Case 1	Manage Candidate Profile
Actor	Candidate
Description	Allows candidates to manage and update their profile details including full name, contact, skills, etc.
Basic Flow	1. Candidates access their profiles. 2. They can update their profiles with new information. 3. After updating, candidates need to save the changes made
Use Case 2	Search Jobs
Actor	Candidate
Description	Allows candidates to find jobs matching the candidate's skills
Basic Flow	1. Candidate accesses the search bar. 2. The candidate enters preferred job field. 3. System displays a list of matching jobs

Use Case 3	View Job Recommendations
Actor	Candidate
Description	Candidates can view a list of jobs that matches the candidate's preferences
Basic Flow	1. Candidate scrolls through recommended jobs. 2. The candidate clicks on a job to view information
Use Case 4	Apply for Job
Actor	Candidate
Description	Allows candidates to submit their resume for the job that they liked or preferred
Basic Flow	1. The candidate prepares a resume. 2. Then, candidate uploads resume to the selected job. 3. After uploading, candidate submits the application
Use Case 5	Browse Job Postings
Actor	Candidate
Description	Candidates will be able to go through or search for posts with detailed information about the job position, salary, working hours, etc.
Basic Flow	1. The candidate selects a job posting from the list. 2. Candidate views detailed information 3. The candidate can decide to apply
Extension	View Job Details: Candidates can click into any job post that is available for detailed understanding of what is required for the job



Use Cases for Employer	
Use Case 1	Manage Job Postings
Actor	Employer
Description	Employers would be able to arrange the posts of the jobs that they will be offering through the platform
Basic Flow	1. The employer creates a post. 2. Employer removes job postings that are no longer available
Use Case 2	Filter/Search Candidates
Actor	Employer
Description	The employers can choose or look for the candidates with the skills and availability that matches most with the job they offered
Basic Flow	1. Employer searches for candidates and applies filters to candidate's skills. 2. System displays best matched candidate. 3. The employer offers job to selected candidate
Use Case 3	View Candidates Recommendations
Actor	Employer
Description	Employers can view a list of suggested candidates that are suitable for the job proposed by the platform
Basic Flow	1. After searching, system displays a list of candidates. 2. Employer reviews recommended candidates
Use Case 4	Browse Candidates
Actor	Employer
Description	The employers would be able to search for the candidates and their information on this platform
Basic Flow	1. Employer selects candidate from the list. 2. Then, the employer views candidate's profile
Extension	View Candidate Details: Allows employers to click into any candidate's profile to view detailed information

Development Model (15%) - Ciara

Chosen Development Methodology

The team at Hire Wire has adopted the Scrum development methodology to assist in the development life cycle. Scrum was chosen as it is ideal for flexible and iterative system design. It accounts for changes to requirements, stakeholder needs, and any other changes that may occur during the development process.

Development Workflow

Scrum uses an iterative and continuous development workflow that breaks development into sprints. To best suit our needs at Hire Wire, a sprint of two weeks was decided. Two weeks is the perfect amount of time to develop, test, collect feedback, and refine requirements before moving on to the next sprint. This allows the team at Hire Wire to adapt to changes and improve the product as we go, saving time and resources in the long run.

Iteration or Task Management Process

To manage tasks using Scrum, user stories, product backlog, and planning are extremely important. During each sprint, the following activities occur:

1. Sprint planning – done at the start of the sprint to plan the development and other work that will take place in the current sprint, as well as identifying the overarching goal of the sprint. Team members are assigned work from the product backlog based on their availability, capacity, roles, and responsibilities.
2. Weekly scrums – weekly meetings that allow team members to voice blockers and potential issues that have arisen during development. These meetings are also used as a way of gaining coverage of what other team members are working on and people's capacity.
3. Development work – team members work on the development of the product corresponding to the work they have been assigned during sprint planning.
4. Sprint review – allows team members and stakeholders to provide feedback to the development that was done during the sprint. Based on the discussions held, the timeline, budget, and other aspects of the product backlog is modified.
5. Sprint retrospective – enables team members to express team processes that worked well or worked poorly. It also identifies how the team worked together/individually and what improvements can be made to processes during the next sprint.

Team Roles and Responsibilities

Role	Person Allocated	Responsibilities
Product Owner	Ciara	- Define features of the product - Decide release date and content - Prioritise features - Adjust features and priority - Accept or reject work results
Scrum Master	Scrum Master designated by Hire Wire	- Represent management - Ensure team is fully functional and productive
Front	James	- Turn designs into functional requirements using CSS and HTML
Design	Siew Tin	- Create wireframes, lo-fi prototypes, and designs for the product
User Experience/UX	Qianran	- Plan out user experience - Connect user needs to business expectations
Back	Nikita	- Makes functional requirements possible using JavaScript and MySQL - Manages website security and database connections

Development Plan (15%) - Siew Tin

Development Plan

This part includes an overview of how the tasks will be conducted and when the tasks will be completed. It also includes the project timeline, milestone planning, task allocation among the members and a Gantt chart for tracking progress.

Project Timeline

The project started on 23rd March 2026. The development process will be conducted in several phases such as planning, designing, coding, error checking and documentation. The current state of the project is in the planning and designing phase, and our team will then proceed to coding phase after deciding a design for the platform.

Week 3 (16/3/26) to week 5 (30/3/26): Planning and designing for the platform interface

Week 6 (6/4/26) to week 9 (4/5/26): Frontend, backend coding, database

Week 10 (11/5/26) to week 12(25/5/26): Error checking on the system and documentation

Development Progress (15%) - Qianran

Development Cycle

The development of the Intelligent Talent Matching Platform follows an iterative development cycle based on Agile principles. The system is developed in stages, allowing continuous improvement and refinement throughout the project lifecycle. Each cycle includes requirement analysis, system design, implementation, testing, and review, ensuring that the system evolves progressively and adapts to changes.

Development Iterations and Completed Tasks The team has successfully completed several crucial early-stage milestones, establishing a solid foundation for the platform:

- Requirement analysis and comprehensive project planning.
- Creation of the project overview and definition of system objectives.
- Identification of core system functionalities and user requirements.
- Development of Use Case Diagrams and initial UML structural designs.
- Initial development of key frontend pages, including job listings and candidate profiles.
- Design and implementation of search and filtering functionalities.

Task Planning, Prioritisation, and Team Allocation

Tasks are planned and prioritized using both Taiga and Trello. Taiga is utilized to manage user stories and product backlog items, while Trello effectively tracks individual task assignments and team progress. Core features—such as candidate profile management, job listing creation, and the matching system—are designated as high-priority tasks and scheduled in the earlier sprints.

Tasks are distributed among team members based on their technical expertise to ensure efficient collaboration and clear responsibility allocation:

- **Ciara Slater (8607497) – Team Leader & Database Administrator:** Responsible for overall project coordination, Agile management, database schema design, and maintaining the project timeline.
- **Qianran Xiu (8788728) – Frontend Developer:** Focuses on core UI architecture, advanced user interactions, candidate registration/profile capabilities, and connecting frontend forms with backend APIs.
- **Siew Tin Lee (9496865) – Frontend Developer:** Responsible for translating Figma UI/UX designs into responsive web pages and refining the visual interface.
- **Nikita Veliev (8773142) – Backend Developer:** Focuses on server-side logic, routing, RESTful API development, and data processing.

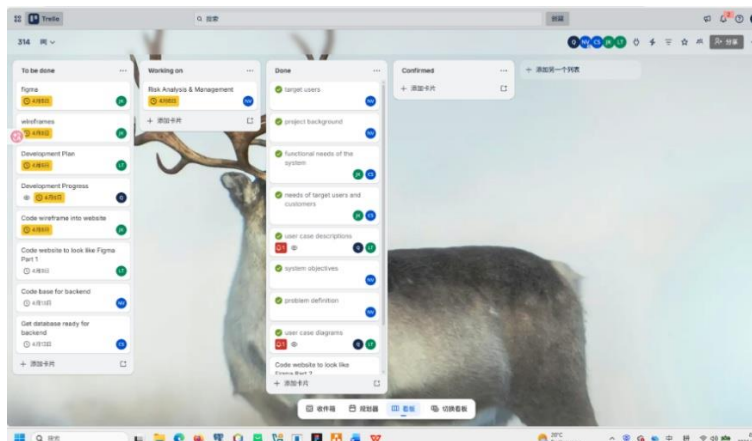
- **James Kennedy (8654086) – Full-Stack Support & UI/UX Design:** Assists with both frontend design implementation and backend integration, bridging the gap between user interface and server functionalities.

Current Progress and Next Steps

At the current stage, the team has successfully transitioned from the design phase into active implementation. Implementation-related tasks, such as coding the interactive website based on Figma designs and backend API preparation, are currently in progress. Advanced features, including the matching/recommendation system algorithm and full frontend-backend integration, are scheduled for the upcoming development stages. Project progress will continue to be monitored transparently via Taiga and Trello to ensure all milestones are met efficiently.

Task Planning and Prioritisation

Tasks are planned and prioritized using both Taiga and Trello. Taiga is used to manage user stories and backlog items, while Trello is used to track individual task assignments and team progress. Core features such as candidate profile management, job listing creation, and the matching system are considered high-priority tasks and are scheduled in earlier stages. Tasks are distributed among team members based on their roles to ensure efficient collaboration and clear responsibility allocation:



Task Progress and Completed Features

At the current stage, the team has successfully completed several major tasks and features, including the design of core system functionalities and UML diagrams. Key features such as job listing creation, candidate profile management, and search/filter functionalities have been designed and partially implemented. Meanwhile, more advanced features such as the recommendation system and backend integration are currently under development and will be completed in the next stages.

Progress Tracking Methods

Project progress is tracked using both Taiga and Trello. Taiga is used to manage backlog items and user stories, allowing the team to monitor overall progress. Tasks are categorized into stages such as "To Do", "Working On", and "Done". Trello is used to track individual task assignments and monitor team member contributions. This helps maintain transparency, improve task management efficiency, and ensure that deadlines are met.

Backlog14 user stories

+ USER STORY

Filters

subject or reference

Tags

USER STORY	STATUS	POINTS	
#1 Create Project Overview	New	7	
#6 Add feature to keep users logged in	New	1.5	
#7 Add reset password feature	New	2	
#8 Create job listings page	New	12	
#9 Add filter search feature to job listings page	New	5	
#10 Matching system	New	7	
#11 Employer job listing upload	New	6.5	
#12 Candidate resume/form upload	New	8	
#13 Employer job listing upload (more basic)	New	7	
#14 Code wireframe	Ready	5	
#15 Code website to look like Figma Part 1	Ready	5	

Risk Analysis & Management (15%) - Nick

This project involves the development of a web-based recruitment platform, which includes a frontend, backend, database, and a recommendation system. The development process involves several risks such as time constraints, technical difficulties, requirement changes, and teamwork issues. These risks can affect the project timeline and overall system quality. Therefore, it is important to identify and evaluate risks before and during the development process and develop strategies to reduce their impact:

Risk	Description	Probability	Impact	Mitigation
Requirement changes	System requirements may change during development	High	High	Use Agile Scrum and flexible design
Time constraints	Limited time to complete the project	High	High	Prioritize core features
Technical difficulties	Issues with backend, database, or integration	Medium	High	Use simple technologies and testing
Team member unavailable	A team member may be unavailable	Medium	Medium	Redistribute tasks among team members
Integration issues	Frontend and backend may not work correctly together	Medium	Medium	Perform regular integration testing
Recommendation system complexity	Matching system may be difficult to implement	Medium	Medium	Use simple matching algorithm
Data loss	Data may be lost due to system errors	Low	High	Regular database backups

Mitigation

To reduce risks, the team will use the Scrum methodology, which allows the team to react flexibly to requirement changes. Tasks will be divided between team members according to their responsibilities. The team will use a simple and reliable technology stack to reduce technical risks. The main system functionality will be developed first to reduce time constraint risks. The team will also perform regular system testing and database backups.

Monitoring

The team will monitor risks throughout the whole project. Project progress will be reviewed during weekly meetings. Tasks will be tracked using Trello. At the end of each sprint, the team will review task completion and overall project progress. If the team faces any issues or falls behind schedule, the team will adjust the development plan.

References

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Technical Document

System Architecture

Provide an overview of the system architecture, including an explanation/description of the way components interact, stuff like that

Frontend Design

Describe the basics of the frontend, stuff like ui/ux and user interaction, as well as the basics of the html and css and js of the system (idk how much detail there needs to be, but assume there should be a decent amount of detail)

Backend Design

Describe how the backend works (how it works with the db, functions, etc), the logic of the system basically, plus the architecture and how the js system works (make references to sections of the code or blocks)

Database

Just explaining the SQL db really, shouldn't take long at all

API Documentation

i'm not sure about what this section should include lol

Testing

Talk about the testing methods used to test the system

CI/CD and Deployment

Talk about the build process and deployment method. IMPORTANT: there needs to be a link to the GitHub repo, showing the full version history

References